



پوهنتون کاردان
KARDAN UNIVERSITY

Database Administration

Lecture 05



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بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ
الْحَمْدُ لِلَّهِ رَبِّ الْعَالَمِينَ
وَالصَّلَاةُ وَالسَّلَامُ عَلَى سَيِّدِنَا مُحَمَّدٍ صَلَّى اللَّهُ عَلَيْهِ وَسَلَّمَ



Overview

Built-in Functions

- Single row character functions
- Case conversion functions
- Manipulation functions
- Examples



Activity

Practice the Character Manipulation Functions



Single Row Functions

- Act on each individual row selected
- Return one result **per row retrieved from table**
- Can be nested
- Function can be used as part of expression in: SELECT clause, WHERE clause
- Argument for function may be any expression, literals or columns.



Single Row Character Functions

CASE CONVERSION FUNCTIONS

LOWER

UPPER

INITCAP



CHARACTER MANIPULATION FUNCTIONS

CONCAT

SUBSTR

LENGTH

INSTR

LPAD, RPAD

TRIM

Case Conversion Functions

- Convert case of character string data
- Useful for matching when unsure of case used for column's data or when case varies by row by row
- UPPER: converts all characters to upper case
- LOWER: converts all characters to lower case
- INITCAP: converts first character of each word to upper case and remaining to lower case

Function	Result
LOWER(' Intro to SQL ')	intro to sql
UPPER(' Intro to SQL ')	INTRO TO SQL
INITCAP(' Intro to SQL ')	Intro To Sql

Case Conversion Examples

Display employee number, name and department number for employees named King

```
SQL> SELECT  employee_id, last_name, department_id
          FROM emp
          WHERE ename = 'smith';
no rows selected
```

```
SQL> SELECT  employee_id, last_name, department_id
          FROM employees
          WHERE LOWER(ename) = 'smith';
```

EMPLOYEE_ID	ENAME	DEPARTMENT_ID
100	SMITH	10



Examples using Character Manipulation Functions

Function	Result
CONCAT(' Good ', ' Day ')	GoodDay
SUBSTR(' Good ',3,2)	od
LENGTH(' Good ')	4
INSTR(' Good ', ' o ')	2
LPAD(last_name,20,'*')	*****King
TRIM('\$' FROM '\$2,345')	2,345



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Thank You